

Step 16 - Mock Interview: 25 Questions

Phase E · Graded Q&A; Drawn From Your Own Build · ~30 min

Goal: The final test - 25 Canadian banking DE questions with model answers drawn from what you actually built. Read each question, answer aloud FIRST, then compare. Artifact: interview-prep/qa.md in your own words.

How to use this

Three passes: (1) answer aloud before reading; (2) mark weak ones, redo next day; (3) have someone fire random questions. Every answer below references YOUR lab - say 'in my project' and offer to show the notebook.

The 25 Questions

Q1. Why explicit schemas instead of inferSchema?

inferSchema drops leading zeros on account numbers, misreads dates as strings, and double-reads the file. StructType is the data contract - mismatches fail loudly.

Q2. Transformations vs actions?

Transformations (filter, select, join) are lazy plan-building; actions (count, write, show) trigger execution, letting Catalyst optimize the whole plan.

Q3. row_number vs rank vs dense_rank?

row_number 1,2,3,4 no ties; rank 1,2,2,4 ties skip; dense_rank 1,2,2,3 no skip.

Q4. rowsBetween vs rangeBetween?

rows counts physical rows; range measures the orderBy value (cast timestamp to long for seconds). 30-day windows need rangeBetween.

Q5. How do you dedup keeping the latest record?

row_number over partitionBy(business key) orderBy(timestamp desc), keep rn=1. dropDuplicates keeps an arbitrary row - not acceptable for financial data.

Q6. When do you use a UDF?

Last resort - UDFs are opaque to Catalyst. Native functions first; I masked SINS with concat + substring, no UDF needed.

Q7. What is a broadcast join?

Ship the small table to every executor to avoid shuffling the big one. I broadcast the 15-row branch dim; the plan shows BroadcastHashJoin.

Q8. Why Parquet over CSV?

Columnar (read only selected columns), 5-10x compression, embedded schema, predicate pushdown. CSV stays for source-in and regulator-out.

Q9. Why Delta over plain Parquet?

ACID writes, time travel for as-of audits, MERGE for upserts and SCD2, schema enforcement, OPTIMIZE/Z-ORDER.

Q10. Explain the Medallion architecture.

Bronze immutable raw + audit columns (restricted - PII in clear); Silver cleansed, deduped, PII-masked single source of truth; Gold business aggregates. Masking at the Bronze-Silver boundary makes downstream PIPEDA-safe by construction.

Q11. How does time travel help OSFI audits?

VERSION AS OF / TIMESTAMP AS OF answers 'state on date X' in one SELECT instead of a backup restore. DESCRIBE HISTORY is the change ledger.

Q12. How do you guarantee exactly-once transaction loads?

MERGE on transaction_id with whenMatchedUpdateAll / whenNotMatchedInsertAll - resends update instead of duplicating; reruns are idempotent.

Q13. Explain SCD Type 2 and how you built it.

Preserve history: expire the old row (is_current=false, end_date=today), insert the new current row. Two Delta MERGE steps; my lab shows a Toronto-to-Mississauga move as two rows.

Q14. What is partition pruning?

Partitioning writes folder-per-value; filtering on that column makes Spark skip entire folders. Visible as PartitionFilters in explain().

Q15. What is the small files problem?

Many tiny files make per-file overhead dominate reads. Fix: Delta OPTIMIZE compaction, Z-ORDER for co-location.

Q16. VACUUM - when and why carefully?

Removes unreferenced files past retention - but kills time travel before that point. In banking, retention is a compliance decision; dry-run first.

Q17. Parameters vs variables in ADF?

Parameters: read-only inputs from trigger/parent/UI. Variables: internal, reassignable via Set Variable.

Q18. How did you design for backfill?

business_date parameter, default empty; an @if picks utcnow() when empty, the passed date otherwise. Proved by rerunning for a past date - two dated folders in bronze.

Q19. How do you authenticate ADF to ADLS in production?

Managed Identity - no secrets, Azure AD-audited, least privilege. Key Vault references when a secret is unavoidable. Account keys are dev-only.

Q20. What is a Self-Hosted Integration Runtime?

ADF's agent installed inside a private network to reach on-prem sources (core banking Oracle) via outbound-only HTTPS - no inbound firewall holes; data stays in-region.

Q21. Name the FINTRAC reports and thresholds.

LCTR cash >= \$10,000; EFTR international transfers >= \$10,000; STR suspicious activity, no threshold. Structuring = staying just below thresholds - my rules flag 9,000-9,999 repeats.

Q22. LCTR vs STR in pipeline terms?

LCTR is mechanical extraction above threshold; STR needs human judgment - the pipeline produces candidates (structuring, velocity), an analyst files the report.

Q23. How do you protect SIDs end to end?

Clear only in restricted Bronze; masked to last-3 at Silver with raw column dropped; all joins downstream use masked or hashed values; masking logic lives in one shared utils notebook for auditability.

Q24. Your EOD batch failed at 3 AM - walk me through recovery.

Alert fires; Monitor shows the failed activity and error; fix or wait out the transient (2 retries already absorbed blips); rerun with business_date=yesterday - safe because loads are idempotent; reconciliation gate confirms totals before Gold rebuilds.

Q25. How do you reconcile the warehouse against the GL?

Join aggregates to GL totals on branch + type + date; three tiers - MATCH (audit proof, kept 7 years), VARIANCE within tolerance (log + notify), FAILURE (halt the EOD via non-zero exit so bad data never reaches FINTRAC reports).

CHECKPOINT — The Finish Line

Track complete when: all 25 answered confidently aloud; qa.md committed in your own words; the 90-second pitch and 3-minute repo walkthrough delivered without reading. You now have what most candidates do not: a public, working, compliance-aware Canadian banking data platform - and the ability to defend every line of it.